

Dhwani Dholaria

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EDUCATION

Texas Tech University

Graduate Student in Master of Science (M.S.) , Computer Science

GPA: 4, Coursework: Analysis of Algorithms, Theory of Automata, Wireless Networks, Image Processing

Lubbock, Tx

August 2023 - May 2025

Pandit Deendayal Energy University

Bachelor of Technology in Computer Engineering

GPA: 3.85, Coursework: Data Structures, Object Oriented Programming, Operating Systems, Computer Architecture

Gujarat, India

July 2019 - May 2022

EXPERIENCE

Texas Tech University

Research Assistant

Lubbock, Tx

September 2024 - Present

- Conducting research on evaluating essays written by humans and edited by ChatGPT-4.
- Calculating metrics to assess quality and improvement, including Flesch Reading Ease, TF-IDF, LDA, lexical diversity, sentiment analysis, cohesion, coherence, and perplexity.
- Analyzing trends in automated and human-written content to provide insights into language models' editing impact.

DI Solutions

Graduate Trainee

Surat, India

December 2022- August 2023

- Led the development of the Redrose website for a premier UK sofa manufacturing company, implementing advanced features such as wood frame ordering, employee management, and barcode generation to enhance the company's digital operations.
- Served as a backend API developer for live projects, including a video chatting platform akin to Zoom, specifically tailored for virtual funeral services, which broadened the company's service offerings.
- Developed numerous RESTful APIs using Node.js, and acquired proficiency in Express.js, facilitating the seamless integration of web applications and services within the organization and Integrated backend APIs seamlessly with the Angular.js frontend to enable dynamic data retrieval and interaction, enhancing user experience and ensuring real-time data synchronization.

Searce Inc.

Software Development Intern

Ahmedabad, India

June 2022 - June 2022

- Gained practical experience with the Google Cloud Platform, focusing on leveraging the Google Cloud API for text-to-speech and speech-to-text functionalities using Python.
- Developed and implemented robust text-to-speech and speech-to-text features within the Google App Engine SDK as part of an internship at Searce Inc., enhancing the company's suite of cloud-based solutions and contributed to optimizing cloud services by integrating advanced API features, which improved the accessibility and interaction capabilities of client applications.

PROJECTS

AI methods for exposing Deepfake videos

- Pre-processed DFDC, FF++ , and Celeb-DF datasets to get the cropped frames containing the face of the speaker.
- Incorporated a traditional temporal feature analysis method, Optical flow to extract the temporal features of the videos.
- Proposed and implemented a hybrid CNN-LSTM model that leverages Optical flow features to identify the authenticity of the videos and classify them as real or fake.

Image Captioning for Hashtag Generation

- Developed an innovative web application for image captioning to generate relevant hashtags for social media posts, significantly enhancing content accessibility.
- Employed Selenium to automate the gathering and processing of image data, ensuring efficient and scalable operations within the application.
- Utilized advanced machine learning techniques for precise image analysis and caption generation, leading to more effective and contextually appropriate hashtag suggestions.

TECHNICAL SKILLS

- Languages/ Frameworks: Javascript, Python, Java, C/C++, Typescript, SQL, HTML/CSS, Node.js, Express.js, React, Bootstrap, OpenCV, Pandas, Excel
- Tools / Platforms: GIT, GitHub, MySQL, SQLite, MongoDB, PostgreSQL, Postman, VS Code

PUBLICATION

- Saikia, P., Dholaria, D., Yadav, P., Patel, V., Roy, M. "A Hybrid CNN-LSTM Model for Video Deepfake Detection by Leveraging Optical Flow Features." *IEEE International Joint Conference on Neural Networks (IJCNN)*, 2022, Padua, Italy.
Developed a novel hybrid CNN-LSTM model that incorporates optical flow-based temporal feature extraction to improve the detection of deepfake videos. Achieved competitive accuracy metrics across multiple datasets (DFDC, FF++, and Celeb-DF).